

Which factors predict susceptibility to phishing? An empirical study

Liliana Ribeiro^{a,*}, Inês Sousa Guedes^{b,c}, Carla Sofia Cardoso^b

^a School of Criminology, University of Porto, Portugal

^b Interdisciplinary Research Centre on Crime, Justice and Security (CJS), School of Criminology, University of Porto, Portugal

^c CEJEA, Lusíada University, Norte, Portugal

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ABSTRACT

Phishing is a cybercrime in active growth that victimizes a large number of individuals and organizations.

To explore which individual and contextual factors predict phishing susceptibility, an online survey was developed, and participants were invited to participate through institutional email from the University of Porto and social networks. The total sample was constituted of 449 individuals.

Results showed that subjects that perceive to have phishing detection self-efficacy and those that have greater use of services in Internet routine activities were more susceptible to phishing. Technology competencies and other individual variables do not predict phishing susceptibility in our sample.

Furthermore, the majority of factors (individual and contextual) tested do not predict phishing susceptibility. So, more studies are needed to understand which factors influence this susceptibility, and regarding that how individuals can protect themselves.

Finally, potential applications of this research include replication in other countries/contexts, and/or the application of the survey together with other innovative tools.

1. Introduction

Phishing is a cybercrime that existed for a long time, but it has received increasing attention in recent years, mainly due to its implications for individuals and organizations. Damages from phishing attacks include social and psychological consequences (Dhamija and Tygar, 2005; Butler, 2007; Tembe et al., 2013), but also financial losses (Hardee et al., 2006; Butler, 2007; Tembe et al., 2013).

According to the FBI's Internet Crime Report in 2021, 323.972 individuals were victimized by phishing attacks, resulting in an estimated \$44.213.000 in damage to victims (FBI, 2021). Moreover, in the UK, Cyber Security Breaches Survey (2022) identified that almost 90 % of UK cyber-attacks directed at businesses were phishing attempts. The victims of these attacks report that phishing is the most disturbing type of attack that they experience, and the costs of attacks represent money or data loss.

As a social engineering attack, this phenomenon can involve different techniques and means (e.g. email, voice calls, social networks), but in general, it involves an attempt to persuade potential victims to take an action that allows offenders to reach their personal information (e.g., Hong, 2012; Khonji et al., 2013; Lastdrager, 2014; Ghazi-Tehrani

and Pontell, 2021). In the Portuguese context, the National center of Cybersecurity's annual report highlights the increase in phishing attacks during 2021 and the fact that phishing emails affect most individuals due to human errors (CNCS, 2022). In fact, despite the existence of some automatic detection of phishing attacks through protective mechanisms, individuals ultimately make the final decision (Xiong et al., 2019). Therefore, phishing susceptibility is understood as the failure to detect phishing emails (e.g., Moody et al., 2017; Vishwanath et al., 2018; Gopavaram et al., 2021). Recently, several studies have been developed to understand which factors predict susceptibility to phishing, asking questions such as "are gender and age predictors of phishing susceptibility?" (e.g., Sheng et al., 2010; Kumaraguru et al., 2009; Hassandoust et al., 2020; Gopavaram et al., 2021), "are technology competencies or training important for correctly detecting phishing?" (Wright and Martet, 2010; Gopavaram et al., 2021; Sarno and Neider, 2021).

Taking into consideration the mixed results that the scientific literature has found when trying to understand the specific factors related to susceptibility to phishing, the current study aims to understand which aspects predict susceptibility to phishing including variables such as gender, age, online activities, and previous cyber-victimization experiences. By using pictures of legitimate and phishing emails presented to

* Corresponding author at: Faculdade de Direito da Universidade do Porto, Rua dos Bragas 223, Porto 4050-123, Portugal.

E-mail address: up201705357@g.uporto.pt (L. Ribeiro).

participants followed by a survey, this study presents an innovative methodology and, to our best knowledge, it is the first in Portugal investigating the factors that predict the susceptibility to phishing among individuals.

In this study, individuals were identified as susceptible to phishing when they failed to identify at least one phishing email and decided to click on the hyperlink present in the email. The paper begins with an overview of the relevant literature, followed by the development of the research hypotheses to be tested. The empirical study is then presented and the results are discussed.

2. Background and literature review

Phishing is a cybercrime that is currently experiencing active growth, and there are several definitions of phishing in the scientific literature (Lastdrager, 2014; Salloum et al., 2022). Although the term ‘phishing’ is widely mentioned (Lastdrager, 2014), it was mainly in the last decade that this term has sparked numerous debates about its definition and scope (Khonji et al., 2013).

Etymologically, the term ‘phishing’ emerged in 1996 through social engineering attacks against America On-Line (AOL). Then, the term was coined as a play on ‘fishing’, as in the fishing activity, where the attackers (in this case, Phishers) use a bait (social engineering message, such as an email) to ‘fish’ (for theft of personal information from victims) (Alabdan, 2020). The transformation of the ‘f’ into ‘pH’ occurred because the first attack known in this area was against a telephone (which was named ‘phone phreaking’) (Khonji et al., 2013).

A phishing attack is a combination of “social engineering” and technical manipulation methods to persuade Internet users “into giving away sensitive information (e.g., using a web form on a spoofed web page) that the attacker can then use to make a financial profit” (Kirda and Krügel, 2006, p. 1). Lastdrager (2014) conducted a study to identify a consensual definition of the phishing term and concluded that “phishing is a scalable act of deception whereby impersonation is used to obtain information from a target.” (p. 8).

More recently, there has been an effort to standardize the term ‘phishing’, based on the premise that phishing is a fraudulent attempt, in social engineering form, using technological and communication means (Welk et al., 2015; Barraclough et al., 2021; Ghazi-Tehrani and Pontell, 2021; Gopavaram et al., 2021). In the present study, we adopt the following phishing definition: a fraudulent method of obtaining personal data (username and respectively password), through email messages, writing messages (SMS or WhatsApp), or telephonic messages. By rule, criminals pose as credible institutions such as Banks, Online Service Providers, etc. with the threat that individuals must take urgent action and upload a link. Thus, misleadingly, individuals provide their personal data, believing that they are inserting it on the website of the real entity.

Furthermore, phishing can be perpetrated through different means, with different motivations for the offenders. Specifically, phishing can be carried out through call voices from offenders towards the victims who believe that the source is legitimate (Vishing). Moreover, offenders can also use SMS or MMS means (‘Smishing’) to conduct their attacks. However, the most common method of launching phishing attacks involves sending a fraudulent email by an offender to victims in order to deceive people into revealing personal data, believing that is a legitimate entity asking for this information. In this sense, three other different types of phishing attacks exist, depending on the motivation behind the email sent by the offender. ‘Spear phishing’ occurs if the email is directed at specific individuals or organizations. If it is directed to higher-status employees or senior people, it is called ‘Whaling’. Finally, ‘Business Email Compromise’, happens when a phishing email is sent to non-profit organizations, commercial and governmental organizations in order to cause financial damage.

Thus, financial benefits, identity theft, industrial espionage, and malware distribution are the principal motivations referred to in the practice of phishing attacks.

3. Explicative models of phishing susceptibility

Some theories and models have been developed to explain phishing susceptibility, failing into two main strands: (1) information processing, which includes the *Heuristic-Systematic Model* (HSM; Chaiken, 1980) and *Suspicion, Cognition, and Automaticity Model* (SCAM; Vishwanath et al., 2018); and (2) the opportunity to act, represented by *Routine Activity Theory* (RAT; Cohen & Felson, 1979).

RAT is the most commonly applied theory in the study of phishing susceptibility, as it is a situational theory that provides insights into the growth of this phenomenon (Ghazi-Tehrani and Pontell, 2021). However, HSM and SCAM argue that information processing plays a crucial role in phishing susceptibility, particularly in the detection of phishing emails. In the following lines, these theories will be briefly outlined.

3.1. Heuristic-systematic model (Chaiken, 1980)

Chaiken (1980) created the Heuristic-Systematic Model (HSM), a sociocognitive perspective that can be considered a ‘double process theory’ because it distinguishes two modes of information processing – systematic and heuristic. Systematic processing, also characterized as deliberate, involves a thorough and analytical treatment of received information, demanding higher cognitive abilities and skills (Chaiken, 1980; Chen and Chaiken, 1999; Luo et al., 2013). On the other hand, heuristic processing is characterized by speed, minimal use of cognitive ability, and a greater reliance on easily recognizable clues, such as format, subject, and message source in an email, to determine its legitimacy. Therefore, individuals use cognitive shortcuts that can lead to quick decisions based on immediate emotions – which can result in distorted decisions due to rapid processing (Chaiken, 1980; Chen and Chaiken, 1999). The evidence shows that individuals are capable of both processes, but there is a prevalence of heuristic over systematic processing (Chen and Chaiken, 1999; Vishwanath et al., 2011). Moreover, the information processing is influenced by contextual factors, such as authority figures (e.g., bank employees, school administrators), social and/or temporal pressures, and personal abilities. When any of these contextual are present, the urge for immediate action increases, leading individuals to rely on heuristic processing (Chaiken, 1987). Specifically, in phishing email detection, an increased susceptibility to failing in correct detection occurs when individuals use heuristic processing because the rapid analysis through cognitive shortcuts limits the recognition of clues in these emails.

3.2. Suspicion, cognition, and automaticity model (Vishwanath et al., 2018)

Vishwanath and colleagues (2018) developed the *Suspicion, Cognition, and Automaticity Model* (SCAM), to explain phishing susceptibility based on previous theoretical assumptions. According to this model, ‘suspicion’ is considered the main endogenous predictor of individual susceptibility to phishing emails (Vishwanath et al., 2018). ‘Suspicion’ is defined as “the degree of uncertainty one has when interacting with a particular stimulus” (Lyons et al., 2011, p. 220). This construct is influenced by the contextual environment, creating discord between an individual’s perceived reality – obtained in a context that influences the evaluation mode and cognitive processing of information – and expected reality – contextually reflecting the individual beliefs about what is plausible (Lyons et al., 2011). Thus, SCAM includes both cognition factors, such as processing modes, and cyber-risk beliefs. Building on HSM, the SCAM postulates that a higher level of heuristic processing will decrease individuals’ suspicion about phishing emails, while systematic processing will increase their suspicion regarding the legitimacy of the same type of email (Vishwanath et al., 2018). In turn, cyber-risk beliefs are related to the individuals’ perceptions of risks associated with online behavior, which serve as a link between individuals’ experience, their effectiveness in dealing with these risks, and the knowledge of each

(idem).

Thus, it is advocated that the higher the cyber-risk beliefs, the lower the heuristic processing, and the higher the systematic processing. Moreover, these beliefs will influence the suspicion of the legitimacy of a phishing email (Vishwanath et al., 2018). If an individual considers the likelihood of cyber-risk to be high, he/she will evaluate the email information more carefully using systematic processing, distrusting its legitimacy, which, in turn, decreases their susceptibility to being victimized by phishing. Researchers have shown that individuals' overconfidence in the correctly detecting phishing has a negative relationship with effective correct detection of phishing (Hong et al., 2013; Wang et al., 2016).

3.3. Routine activity theory (Cohen & Felson, 1979)

Cohen and Felson's (1979) Routine Activity Theory (RAT) is the most widely applied "traditional" criminological theory in the field of cybercrime (Yar, 2005; Leukfeldt, 2015; Leukfeldt and Yar, 2016; Ghazi-Tehrani and Pontell, 2021). Their premise is that crime will occur when there is a convergence in time and space of a motivated offender, incapable guardians (e.g., police, surveillance cameras, people), and a suitable target (determined by Value, Inertia, Visibility, and Access – VIVA). The 'value' can vary depending on the crime, but in a phishing attack, it relates to motivations such as financial gain. Therefore, individuals with higher incomes may be at a greater risk of receiving phishing emails (Graham and Triplett, 2017). In the cyberspace context, 'visibility' might be facilitated due to the lack of physical barriers on the Internet, which in turn may attract more motivated offenders. (Yar, 2005). Furthermore, 'inertia' applies to people or objects, which, in cyberspace context, could be exemplified when an attachment is downloaded with a virus (Leukfeldt and Yar, 2016). Finally, 'access' can be limited by the potential victims by implementing various security measures, such as using strong passwords.

However, there is a debate about the application of RAT's theory in the phishing field. Yar (2005) argues that temporal convergence between the victim and offender does not exist in a phishing attack, while Ghazi-Tehrani and Pontell (2021) advocate that this convergence can be understood broadly as temporal contact, for example, between email phishing and the victim. Thus, phishing susceptibility may be lower when a 'suitable target' is less exposed on the Internet or when using cybersecurity mechanisms, such as an updated Internet browser.

4. Empirical studies on phishing

The study of the predictors of phishing susceptibility has been increasing in response to the corresponding growth in the visibility of phishing attacks (Williams et al., 2017). Several researchers have been trying to understand the factors that predict phishing susceptibility, including variables such as gender, age, education, phishing knowledge, and online activities. However, the results of these studies have been mixed. In the following lines, empirical data on the relationship between phishing susceptibility and a set of individual and contextual variables will be described.

4.1. Gender, age, computer training, and skills

Concerning the relationship between gender and phishing susceptibility, while some studies have shown a relationship between them (Sheng et al., 2010; Hong et al., 2013; Abroshan et al., 2021), others indicate that this relationship is non-existent (Canfield et al., 2016; Sarno et al., 2017; Gopavaram et al., 2017).

Specifically, authors such as Sheng and colleagues (2010), Hong and colleagues (2013), and Abroshan and colleagues (2021) observed that women were more susceptible to phishing than men, failing more in identifying phishing emails (Sheng et al., 2010; Hong et al., 2013). One possible explanation is that women have less motivation or fewer

opportunities to learn about phishing attacks (Sheng et al., 2010). Other factors can influence this susceptibility (Abroshan et al., 2021), such as women's Internet use or their tendency to follow instructions more promptly (Sheng et al., 2010). Nevertheless, some studies showed that gender does not predict susceptibility to phishing, regardless of the methodology used (e.g., Sarno et al., 2017; Albladi and Weir, 2018; Parsons et al., 2019; Verkijika, 2019). For example, Gopavaram et al. (2021), in a transnational study including five different countries (Australia, Canada, New Zealand, England, and the USA) concluded that gender did not predict higher or lower susceptibility to phishing.

In addition, mixed results concerning the relationship between phishing susceptibility and age have been found. Darwish et al. (2013, p. 4) argued that age is an important predictor of this phenomenon, stating that "older individuals were less likely to fall prey for phishing; while younger individuals were consistently more vulnerable to phishing attacks". However, other studies refute this prediction (Flores et al., 2015; Moody et al., 2017; Sarno et al., 2017). Authors such as Kumaraguru and colleagues (2009) and also Sheng and colleagues (2010) concluded that younger individuals (between 18 and 25 years old) had more susceptibility to phishing than older individuals (26 years or older). Nevertheless, recent studies found opposite results, concluding that older individuals are more susceptible (e.g., Gopavaram et al., 2021; O'Connor et al., 2021). For example, Lin and colleagues (2019) through a study undertaken in Florida, concluded that there was a high susceptibility to phishing and less awareness of it in older individuals. The specificities of the methodologies used in different studies, namely the existence (or not) of a variety of phishing emails by researchers, is one explanation given for the mixed results (Lin et al., 2019).

Interestingly, when integrating gender and age, authors observed that older women report higher susceptibility to phishing since women present lower knowledge and lower computer and Internet skills than men (Lin et al., 2019). Currently, computer knowledge is considered crucial for better phishing email identification (Gopavaram et al., 2021). In fact, education seems fundamental in this area (Jakobsson and Menczer, 2007; Albladi et al., 2018), as it is considered the best way to prevent and protect against phishing attempts (Wright and Marett, 2010; Miyamoto et al., 2014; Ghanzi-Therani & Pontell, 2021). Researchers have suggested a negative relationship between education and phishing susceptibility – the more educated the individuals, the smaller their susceptibility to this phenomenon is (Moody et al., 2017). In addition, previous studies found that higher levels of knowledge about the online environment are related to an increased perception of threats in emails and a greater perception of the difficulty of correctly distinguishing legitimate from phishing emails (Downs et al., 2007; Sarno and Neider, 2021).

Finally, despite the scientific evidence that the Internet experience leads to a decrease in phishing susceptibility (Wright and Marett, 2010), recent research argues that individuals who used Internet more frequently, in general, are more prone to phishing susceptibility, perhaps because of their willingness to click on the links in received emails (Moody et al., 2017). Interestingly, in a study undertaken by Albladi and Weir (2018), male experts disagree that Internet use influences the susceptibility to phishing, while female experts consider it a very important factor, especially for the perception of perceived risk and severity, which may influence susceptibility to phishing.

4.2. Previous experience of victimization

The influence of previous experiences of victimization by phishing attacks has been studied in some research. Still, studies that include this variable focused on a single situation or on knowing the susceptibility between situations. In the latter case, they argue that victims learn from experience, which influences susceptibility at the time of analysis (Chen et al., 2020).

One of the factors that contributes to individuals learning from previous experiences is the knowledge of phishing emails. Being

victimized or being alerted by the existence of a particular email reduces their probability of victimization (Downs et al., 2006). In fact, although there is some evidence that previous experience of victimization does not decrease phishing susceptibility (Hong et al., 2013), the majority of studies agree that this experience works as an educational tool (Wang et al., 2012). For instance, Canham et al. (2019) argue that a small percentage of individuals are 'Repeat Clickers', and this behavior also can be influenced by various factors, including individual and situational.

Additionally, the existence of a monetary loss derived from responding to a phishing email, without the amount being refunded, motivates individuals to adopt more cautious behaviors in the future, reducing susceptibility to phishing (Welk et al., 2015). Therefore, previous experience of victimization increases the adoption of security practices to protect against the phenomenon (Chen et al., 2020; Hassandoust et al., 2020), namely the greater probability of considering emails as illegitimate (O'Connor et al., 2021).

4.3. Internet routine activities

The RAT is one of the most tested theories in the field of cybercrime, but few studies have explored the Internet's routine activities in relation to phishing victimization risk (Holt and Bossler, 2013).

Concerning the relationship between phishing victimization risk and Internet routine activities, mixed results have been found. While some studies have pointed out a relation between the greater use of certain Internet routine activities and phishing victimization risk (e.g., Reyns et al., 2011; Reyns, 2015), others did not find such a relationship (e.g., Ngo and Paternoster, 2011; Leukfeldt, 2014). In fact, some evidence suggests that the more time individuals spend on the Internet, the higher their exposure to phishing attacks (Reyns et al., 2011). To explore the effects of several routine activities on three types of offenses, including phishing, Reyns (2015) conducted a study with a total sample of 10,925 participants. The results suggested that online routine activities such as online banking, online booking, online purchasing, online social networking, regularly changing passwords, and deleting emails had a positive relationship with the risk of phishing victimization. In turn, posting accurate information online showed a negative relationship with the dependent variable of the study (Reyns, 2015), indicating that more information sharing leads to less phishing victimization.

Another relevant study (Leukfeldt, 2014) with a total sample of 10,316 individuals, aimed to understand which factors lead to an increased risk of victimization and concluded that only surfing the Internet increased phishing victimization risk. No other activities showed a relationship with phishing susceptibility. Ngo and Paternoster (2011), using a sample of 295 individuals, found that spending time online with specific activities did not increase phishing victimization risk (Ngo and Paternoster, 2011). This result supports the evidence of Holt and Bossler's study (2009) that Internet activities such as spending the most time online, writing emails, and opening dubious hyperlinks do not increase phishing victimization risk.

Taking into consideration these mixed results, further studies are needed to understand the influence of Internet routine activities on phishing victimization risk.

4.4. Current study

This empirical study aims to explore susceptibility to phishing through the administration of a self-reported survey. Specifically, it aims to: (a) understand if emails are detected differently according to individual characteristics – gender, age, education, and monthly familiar income; (b) understand whether the perception of self-efficacy in detecting emails and the perception of phishing severity influences susceptibility to phishing; (c) explore if (and which) Internet routine activities influence susceptibility to phishing; and (d) understand whether individual differences that can predict susceptibility to

phishing are influenced by previous cyber-victimization experience, fear and/or perceived cyber-victimization risk of a phishing email. In order to ensure uniform comprehension of the term phishing among all participants and to diminish potential ambiguities in understanding, a specific definition of phishing, already described in the Literature Review section, was incorporated into the questionnaire.

4.5. Methodology

4.5.1. Data

The data for this research were collected during February and March 2022 through an online survey, which was developed to achieve the aims of the present research. After obtaining Institutional authorization, an email containing the study's objectives and the survey link was sent out through the University of Porto, inviting both staff and students to participate. Additionally, the survey link was also disseminated on social networks (e.g., Facebook, LinkedIn). The total sample was constituted of 449 individuals, with the majority being women (69.6 %).

4.5.2. Measures

In the next lines, the operationalization of the dependent and independent measures will be described.

4.5.2.1. Dependent variable. In the present study, the dependent variable was considered: phishing susceptibility. To operationalize this variable, four emails (two legitimate emails and two phishing emails), were presented to individuals in the survey, followed by a question asking "Would you open the hyperlink present in the email?". Answers varied between "Yes" and "No". Furthermore, depending on the answer, they were forwarded to the question of reasons to open (or not) this hyperlink (e.g., knowledge of the institution/entity that sends the email). Individuals were considered susceptible to phishing when they fail in at least one correct phishing identification, which occurred when they choose to open hyperlinks in the phishing email.

4.5.2.2. Independent variables. The independent variables included individual and contextual factors: sociodemographic variables, Internet routine activities, phishing knowledge, perception of phishing severity, phishing detection self-efficacy, fear of identity theft, perceived cyber-victimization risk, previous phishing victimization, cyber-victimization (general).

Regarding sociodemographic variables, five individual characteristics were included: (a) gender (1 = male; 2 = female), (b) age (in years), (c) education (1 = up to 9th, 4 = post-graduation), and (d) monthly familiar income (1 = less than 250 euros; 7 = more than 1501 euros). These four variables were based on Sheng and colleagues (2010), Canfield and colleagues (2016), Guedes et al. (2022), Couto (2019), Oliveira (2019), and Pfeffel et al. (2019) studies.

Concerning the perception of technology competencies, individuals were asked about their degree of agreement with a set of 22 affirmations, answering on a five-points Likert scale (1 = strongly disagree; 5 = strongly agree). For example, being able to "Add and delete information from a data file.", "Solve problems in computer", "Detect hardware problems of a computer", and "Change and write computer programming" (item 17). The internal consistency of the scale was high ($\alpha = 0.89$).

To examine Internet routine activities, the frequency of specific online routines was rated by individuals on a scale of 1 (never) to 7 (only at weekends). Following Guedes and colleagues (2022) and Reyns (2013), individuals were asked "In a typical week, how often do you use the Internet for the ensuing purposes?": (1) access to online banking; (2) access to email; (3) utilize social media networks (to comment, visualize and share); (4) communicate with family and/or friends; (5) download documents, videos, music, etc.; (6) watch videos, films and/or series; (7) online play; (8) read journals, magazines and/or books; (9) online

shopping; (10) work; (11) access to State services (e.g., health services). The alpha coefficient was 0.702.

After performing a factor analysis, the items were aggregated into four main routine activities (1) services activities, (2) leisure activities, (3) work and email activities, and (4) social activities. This was based on a previous study by Guedes and colleagues (2022).

Following Perrault (2017), ‘phishing knowledge’ was operationalized by asking individuals if they had previous knowledge of the phenomenon (Yes/No), immediately after a presentation of the phishing definition. Also based on Perrault’s study (2017), individuals were asked about their degree of agreement with three affirmations: (1) *phishing is a serious threat*, (2) *phishing is harmful*, and (3) *phishing is a severe issue*, allowing the operationalization of the perception of phishing severity variable. The internal consistency obtained was moderately high ($\alpha = 0.79$).

To operationalize phishing detection self-efficacy, two affirmations were presented to be evaluated: (1) I am able to spot phishing emails, (2) I am able to distinguish phishing from legitimate email (Williams and Joinson, 2020). Individuals rated the two sets of affirmations on a five-points Likert scale varying between 1 = Strongly Disagree and 5 = Strongly Agree. The alpha coefficient was high ($\alpha = 0.91$).

Moreover, to measure the perceived cyber-victimization risk, following Martins (2022), each individual was asked about the likelihood of three situations happening: (1) Someone steals your personal and financial data online, (2) Someone utilizes your personal and financial data online to reach financial benefit, (3) Someone damages your reputation through illegitimate utilization of your personal and financial data online. The individuals answered on a five-point Likert scale where 1 = Not at all and 5 = Very Much. The alpha coefficient was 0.89.

Departing from one of the aims of the present study, analyzing the influence of fear and previous victimization experiences of phishing, individuals were asked a set of questions. Regarding fear of identity theft, based on Neves’s (2022) study, individuals were exposed to the same affirmations previously mentioned in perceived cyber-victimization risk operationalization and rated on a four-point Likert scale where 1 = Not afraid and 4 = Very afraid. The internal consistency of the scale was high ($\alpha = 0.92$).

Previous cyber-victimization experiences were measured through a set of ten questions: “In the past 12 months, how many times”: (1) Did you receive hostile or aggressive messages which created harm or discomfort to you through the Internet or other electronic device?, (2) Did someone, repeatedly and intentionally, impose any type of unwanted communication, proximation or persecution through the Internet or other electronic device?, (3) Did someone threaten to reveal online the information about you if you did not realize one action such as a payment?, (4) Did someone appropriate or use your personal or financial data for criminal purposes, without your consent?, (5) Did someone use your personal data to create a fake profile, without your consent?, (6) Did someone try to access your electronic devices, without an authorization?, (7) Did you receive fraudulent emails or messages asking your personal information?, (8) Did you find malicious software in your device (e.g., virus, spyware)?, (9) Did you buy products and services on the Internet which did not arrive in your home or arrived but were counterfeit products or different of announced?, and (10) Were you a fraud victim in some purchase or sale through MB WAY?. The participants answered between 0 (zero) and 4 (four) or more. The alpha coefficient was 0.86. These questions were based on previous studies (Martins, 2022) and allowed knowing the previous cyber-victimization in general, and the phishing victimization in specific.

4.5.3. Data analysis

Several analyses were performed on the data. Descriptive analysis was conducted to describe the sociodemographic characteristics of participants, while inferential analysis was carried out to explore the relationship between variables.

Kolmogorov–Smirnov normality tests to assess the normality of the data were conducted. The results showed that data did not follow a normal distribution. Therefore, non-parametric tests were employed. To analyze the difference between susceptible and non-susceptible individuals to phishing, as well as between potential victims and non-victims of phishing attacks, Mann–Whitney U tests were executed. Despite the use of Mann–Whitney tests, for comparative purposes mean levels of each variable was presented in tables. The choice to show the means in the tables does not undermine the validity of the statistical analysis, but rather serves to enhance the clarity of the results.

Concerning phishing susceptibility, the dependent variable, considering its dichotomous nature (0 = no; 1 = yes), binary logistic regressions were performed, incorporating both individual and contextual variables to examine their predictive effects.

5. Results

Table 1 presents the descriptive results concerning the sociodemographic variables. Approximately 70 % of the sample was composed by women with a mean age of approximately 29 years old. Regarding education level, 39.6 % of individuals reported having up to 12th grade, and 35 % of the individuals reported having a bachelor’s degree. Moreover, a majority of individuals were students (52.3 %), and while their monthly familiar income varied, the most commonly reported income range was above 1500 euros (28.5 %). Finally, participants reported having intermediate technology competencies ($M = 3.63$; $SD = 0.52$), on a scale with a maximum score of five.

5.1. Differences between susceptible and non-susceptible individuals

Table 2 presents the mean levels of the key variables of this study. Concretely, it shows the means for the total sample and the difference between two subsamples: individuals susceptible and non-susceptible to phishing. Mann–Whitney U tests were conducted in order to assess the differences in susceptibility to phishing emails. The results indicate significant differences exist between susceptible and not susceptible individuals to phishing in terms of technology competencies ($p < .001$), online routine activities related to leisure ($p = .019$), fear of identity theft ($p < .001$), perceived cyber-victimization risk ($p = .038$), and phishing detection self-efficacy ($p < .001$). Specifically, non-susceptible individuals exhibit higher levels of technological competencies, engage

Table 1
Descriptive results of the sociodemographic variables.

Variable	% (N)	M±SD	(N = 449) Min–Max
Age		28.69 ± 12.03	18–75
Gender (Female)	69.7 (313)		
Education			
Up to 9th grade	2.8 (13)		
Up to 12th grade	39.6 (178)		
Bachelor degree	35.7 (160)		
Postgraduation	21.9 (98)		
Current Professional Status			
Employed	32.1 (144)		
Unemployed	3.3 (15)		
Student	52.3 (235)		
Other	12.3 (55)		
Monthly Familiar Income (euros)			
<250	8.2 (37)		
251–500	4.9 (22)		
501–750	10.5 (47)		
751–1000	17.6 (79)		
1001–1250	1.6 (70)		
1251–1500	14.7 (66)		
>1500	28.5 (128)		
Technology Competencies		3.63 ± 0.52	1.59–5.00

M, Mean; SD, standard deviation; Min, Minimum; Max, Maximum.

Table 2

Differences between susceptible and non-susceptible individuals to Phishing emails.

	Total sample		Susceptible (n = 70)			Non-susceptible (n = 379)			(N = 449)	
	M±SD	Min–Max	M±SD	Mo	Mdn	M±SD	Mo	Mdn	U	Effect size
Technology Competencies	3.63 ± 0.52	1.59–5.0	3.46 ± 0.46***	3.67	3.39	3.66 ± 0.53***	3.67	3.67	9978.50	0.16
Internet routine activities										
Services	2.05 ± 0.98	0–5	2.22 ± 0.98	1.33	2.00	2.02 ± 0.98	2.00	2.00	12,203.00	0.05
Leisure activities	2.75 ± 1.05	0–5	2.50 ± 1.06*	2.00	2.50	2.80 ± 1.04*	2.75	2.75	10,939.50	0.11
Work and email	4.17 ± 1.09	0–5	4.02 ± 1.14	5.00	4.50	4.20 ± 1.08	5.00	4.50	12,031.50	0.06
Social activities	4.48 ± 0.92	0–5	4.55 ± 0.91	5.00	5.00	4.47 ± 0.93	5.00	5.00	12,521.00	0.04
Perception of Phishing Severity	4.63 ± 0.55	1–5	4.57 ± 0.49	5.00	3.67	4.64 ± 0.56	5.00	5.00	11,673.50	0.08
Phishing detection self-efficacy	3.78 ± 0.80	1–5	3.31 ± 1.00***	4.00	3.25	3.87 ± 0.73***	4.00	4.00	8714.50	0.23
Fear of Identity Theft	3.28 ± 0.76	1–4	3.54 ± 0.68***	4.00	4.00	3.23 ± 0.77***	4.00	3.33	9680.50	0.18
Perceived Cyber-victimization Risk	2.45 ± 0.73	1–5	2.62 ± 0.84*	2.00	2.50	2.42 ± 0.70*	2.00	2.00	11,263.00	0.10
Previous Phishing Victimization (Frequency)	1.93 ± 0.81	1–3	1.90 ± 0.89	.00	3.00	1.94 ± 0.80	.00	2.00	11,622.00	0.08

The values in bold reached statistical significance.

M, Mean; SD, standard deviation; Mo, Mode; Mdn, Median; * $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$ (p -values were calculated using the Mann–Whitney test); U, value of Mann–Whitney U test.

in more leisure-related online routine activities, and have higher levels of phishing detection self-efficacy compared to susceptible individuals. Additionally, susceptible individuals demonstrate higher levels of fear of identity theft and a greater perceived cyber-victimization risk than non-susceptible individuals. No further statistically significant differences were found among the variables analyzed. Furthermore, the effect size values are below 0.20, indicating low effects, except for the effect size related to the Phishing detection self-efficacy variable.

5.2. Logistic regression on the susceptibility to phishing

To understand the individual and contextual factors that influence susceptibility to phishing, a logistic regression model was performed (Table 3). The model reached statistical significance ($p = .025$) and explains 20.9 % of the variation in the dependent variable. Among the independent variables, only two reached statistical significance. First, phishing detection self-efficacy showed a significant effect on susceptibility to phishing. Individuals with higher levels of phishing detection self-efficacy were significantly less susceptible to phishing than those with lower levels of phishing detection self-efficacy (OR = 0.525; $p = .040$). Secondly, the frequency of Internet activities related to services (such as banks) had a significant impact on susceptibility to phishing. Greater use of services increased susceptibility to phishing (OR = 1.924; $p = .015$).

None of the other variables, whether individual or contextual, were found to be predictors of susceptibility to phishing in this study. Further examination using the Hosmer–Lemeshow test were employed, showing that the logistic model demonstrated a good fit to the data ($p = .858$).

5.3. Previous phishing victimization attempt

Considering the assumption that previous victimization has not been widely studied and that a majority of studies focus on phishing susceptibility (the intentionality of individuals to click or not in hyperlinks), this section explores the characteristics of potential victims of phishing attempts (i.e., individuals who received phishing attacks to ask for their personal data). Specifically, our participants indicated how many phishing emails they received in the last year, which allowed us to determine the victimization attempt. We aim to understand whether potential victims and non-potential victims differ from each other using the same variables tested in differences between susceptible and non-susceptible individuals to phishing.

Table 4 presents the mean levels of the two subsamples: potential victims and non-potential victims of phishing. Non-parametric Mann–Whitney U tests were conducted to assess the differences in studied variables. The results show significant differences between potential victims and non-potential victims in five variables: routine activities

Table 3

Logistic regression: Phishing susceptibility.

Predictors variables	B	SE	Exp (B)	95 % CI	
				Lower bound	Upper bound
Constant	1.02	3.23	2.76		
Age	.010	.022	1.01	.97	1.06
Gender (Male)	.95	.65	2.58	.73	9.18
Technology Competencies	−0.12	.56	.89	.30	2.64
Education (University Education – Yes)	−0.59	.47	.56	.22	1.39
Monthly Income Family (euros)	−0.07	.13	.93	.72	1.20
Internet routine activities					
Services	.66	.27	1.92*	1.14	3.26
Leisure activities	−0.48	.27	.62	.37	1.05
Work and email	−0.24	.27	.79	.46	1.33
Social activities	.24	.32	1.27	.68	2.35
Phishing knowledge	−0.05	.61	.67	.20	2.22
Perception of Phishing severity	.00	.45	1.00	.41	2.43
Phishing detection self-efficacy	−0.64	.31	.53*	.28	.97
Cyber-victimization (General)	−0.01	.06	.99	.88	1.12
Phishing victimization (Frequency)	−0.20	.30	.82	.46	1.47
Model X²	26.125				
Df	14				
P	.025				
Nagelkerke R²	.209				
Hosmer–Lemeshow test	Chi-Square	Df	p		
	3.005	8	.858		

The values in bold reached statistical significance.

* Correlation is significant at the 0.05 level (2-tailed); ** Correlation is significant at the 0.01 level (2-tailed); *** Correlation is significant at the 0.001 level (2-tailed); B, unstandardized beta; SE, standard error; Exp (B), exponentiation of the B coefficient.

related to work and email ($p < .001$), perception of phishing severity ($p = .046$), phishing detection self-efficacy ($p < .001$), perceived cyber-victimization risk ($p = .029$) and general cyber-victimization ($p < .001$). Specifically, potential victims of phishing attacks engage in more Internet activities related to work and email than do non-potential victims. Additionally, potential victims have a higher perception of phishing severity and phishing detection self-efficacy compared to non-potential victims. Finally, potential victims report a higher overall experience of cyber-victimization and a higher perceived cyber-

Table 4Differences between potential victims and non-potential victims of Phishing attempts ($N = 449$).

	Potential victims ($n = 347$)			Non-potential victims ($n = 102$)			<i>U</i>	Effect size
	Mo	Mdn	$M \pm SD$	Mo	Mdn	$M \pm SD$		
Technology competencies	3.67	3.67	3.04 ± 0.72	3.81	3.57	2.96 ± 0.67	16,302.00	0.06
Internet activities								
Services	2.00	2.00	2.09 ± 0.95	1.33	2.00	1.92 ± 1.07	15,611.50	0.09
Leisure activities	2.75	2.75	2.78 ± 1.03	2.50	2.50	2.65 ± 1.10	16,112.50	0.07
Work and email	5.00	4.50	$4.28 \pm 0.98^{***}$	5.00	4.00	$3.79 \pm 1.33^{***}$	14,081.50	0.16
Social activities	5.00	5.00	4.49 ± 0.91	5.00	5.00	4.46 ± 0.95	17,287.00	0.02
Perception of Phishing Severity	5.00	5.00	$4.66 \pm 0.51^*$	5.00	4.67	$4.51 \pm 0.67^*$	15,598.50	0.09
Phishing detection self-efficacy	4.00	4.00	$3.86 \pm 0.72^{***}$	4.00	3.75	$3.51 \pm 0.97^{***}$	13,921.50	0.17
Fear of Identity Theft	4.00	3.33	3.26 ± 0.76	4.00	3.67	3.33 ± 0.79	16,389.00	0.06
Perceived Cyber-victimization Risk	2.00	2.33	$2.50 \pm 0.72^*$	2.00	2.00	$2.31 \pm 0.71^*$	15,273.00	0.10
Cyber-victimization (General)	3.00	3.00	$3.77 \pm 3.84^{***}$	.00	.00	$1.18 \pm 1.72^{***}$	7795.50	0.40

The values in bold reached statistical significance.

M, Mean; SD, standard deviation; Mo, Mode; Mdn, Median; * $p \leq .05$; ** $p \leq .01$; *** $p \leq .001$ (p -values were calculated using the Mann–Whitney test); U , value of Mann–Whitney U test.

victimization risk than non-potential victims. No statistically significant differences were found for the other variables.

In this analysis, it is also noted that the only effect size with a moderate value (0.40) pertains to the Cyber-victimization (General) variable, with all the others having values below 0.20.

6. Discussion

The present study aimed to understand the influence of individual and contextual factors on phishing susceptibility. The results, based on a sample of 449 individuals, suggest that only a few of the tested variables predict phishing susceptibility as shown on Fig. 1.

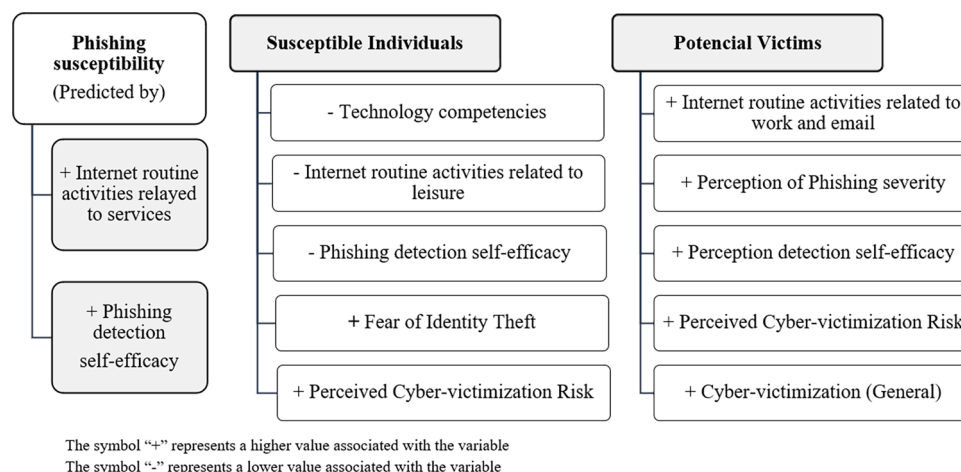
This research did not find a significant difference between men and women in phishing susceptibility. Specifically, in line with previous studies such as Albladi and Weir (2018), and Gopavaram et al. (2021), gender was not a predictor of phishing susceptibility. This result contradicts the finding of Sheng et al. (2010), who found that women were more susceptible to phishing than men. Similarly, age was also not a predictor of phishing susceptibility, in the present study. While some previous researches have suggested that younger individuals are more susceptible to phishing compared to older individuals (Kumaraguru et al., 2009; Sheng et al., 2010), other studies have indicated the opposite, with younger individuals being less susceptible (Lin et al., 2019; Gopavaram et al., 2021; O'Connor et al., 2021). Taking into consideration the results of this study, some explanations can be proposed. The slight differences in the methodology used in the present study compared to other studies, like Sheng and colleagues (2010), may contribute to the differences in findings. Furthermore, the high

percentage of women and young adults, in this research, may have influenced the final results. To achieve more robust findings, it is essential for future studies to strive for a more balanced sample in terms of gender and age representation. Thus, replicating this study in other contexts will be important to support or challenge these results and provide a more comprehensive understanding of the relationship between gender, age, and phishing susceptibility.

The results of this research indicate that higher levels of technological competencies do offer protection against phishing susceptibility. This means that individuals with basic competencies such as simply opening and closing files, are significantly different in their susceptibility to phishing compared to those with more advanced competencies skills, like programming. These findings are according to the evidence found by Wright and Marett (2010) and can be explained by the processing mode used to distinguish phishing from legitimate emails (Chaiken, 1987; Vishwanath et al., 2018).

Concerning the influence of phishing knowledge and the perception of phishing severity, these variables did not predict phishing susceptibility in this study. This finding is contrary to other studies that suggest a greater perception of the severity of the threat may lead individuals to seek more information and adopt protective behaviors against phishing (e.g., Arachchilage and Love, 2013; Hassandoust et al., 2020; Williams and Joinson, 2020). The lack of significance in this study may be attributed to the specific sample or the measures used to assess these variables.

Interestingly, phishing self-efficacy emerged as the only significant predictor of phishing susceptibility. This result can be explained by the higher self-confidence of individuals, which motivates them to pay

**Fig. 1.** Summary of results.

attention to cues in emails, enabling them to distinguish phishing attempts more effectively. The theoretical frameworks of the HSM (Chaiken, 1987) and the SCAM (Vishwanath et al., 2018) support these findings. According to these models, individuals with higher cyber-risk beliefs and self-efficacy are more likely to process email information systematically, leading to a decrease in phishing susceptibility.

Regarding the relationship between Internet routine activities and phishing susceptibility, the study found that a greater use of the Internet for activities related to leisure decreases phishing susceptibility. This finding is contrary with Reyns (2015), who reported a positive relationship between online banking and phishing victimization risk, and it aligns with other studies that did not find a relationship between specific online routine activities (e.g., writing emails and opening stranger hyperlinks) and phishing susceptibility (Holt and Bossler, 2009; Ngo and Paternoster, 2011). Therefore, we can conclude that the results of this research refute the assumptions of the RAT (Cohen & Felson, 1979), which suggests that the more individuals use certain services (e.g. banks) online, the higher their exposure (suitable target) and, consequently, the higher their phishing susceptibility. Moreover, greater use of these activities may lead individuals to process emails using heuristic processing, thereby increasing their phishing susceptibility (Chaiken, 1987; Vishwanath et al., 2018).

Interestingly, former cyber-victimization experiences and previous phishing victimization did not predictor phishing susceptibility in this study. This finding contrast with the study by Hong et al. (2013) where they argued that previous cyber-victimization experience did not decrease phishing susceptibility. Furthermore, it is contrary to studies that mention that individuals with previous cyber-victimization experiences tend to adopt more cautious behaviors and, therefore, are less susceptible to phishing (Welk et al., 2015; Chen et al., 2020; Hassandoust et al., 2020). Chan-Tin et al. (2022) also suggested that other factors can influence the victimization of catphishing, such as having a stronger systematic decision-making style. Moreover, self-control could be an important factor (which we did not test) to explain these findings. In fact, authors like Pratt (2016) have suggested differences in victimization between individuals with high and low self-control. Similarly, studies have shown that higher levels of self-control (specifically lower impulsivity) decrease phishing susceptibility (e.g., Kumaraguru et al., 2007; Pattinson et al., 2012; Mayhorn et al., 2015; Neupane et al., 2016; Tjostheim and Waterworth, 2020). Finally, we can hypothesize that individuals who are overconfident in their perception of self-efficacy in detecting phishing attempts might be more prone to phishing victimization.

Despite this, our results show that potential victims and non-potential victims of phishing are statistically different in several aspects, including Internet routine activities related to work and email use, perception of phishing severity, phishing detection self-efficacy, and previous general cyber-victimization experience. Specifically, potential victims scored higher on these variables compared to non-potential victims. To explain these findings, Wang and colleagues (2016) argued that the overconfidence of individuals in their capacity to correctly identify phishing emails may lead to a bias, which requires further investigation to understand its influence on phishing susceptibility. Concerning our findings that suggest a greater use of the Internet for work and email use, some explanations can be proposed. For instance, the quantity of received and sent emails on a daily or weekly basis may lead to incorrect phishing identification by individuals. In fact, some evidence suggests that a higher volume of emails is associated with a lower ability to correctly identify phishing emails (Sarno et al., 2017; Sarno and Neider, 2021). Moreover, this high volume of emails can lead individuals to read and react to them more automatically, without engaging in systematic processing and paying proper attention to cues in phishing emails.

7. Conclusion and limitations

The growth of phishing highlights the importance of our study and the relevance of identifying which factors predict susceptibility to phishing.

This study holds significant implications for the scientific literature, particularly the finding that the commonly studied individual variables such as gender and age, are not predictors of phishing susceptibility. With these variables being less relevant, it becomes crucial to explore other factors that influence this phenomenon. Moreover, contextual variables, such as greater use of services, have proven their importance in phishing susceptibility, but not in phishing victimization, where work and email activities appear to be more relevant. Understanding the reasons for this difference between victims and susceptible individuals in their Internet routine activities should be the focus of future studies.

Furthermore, phishing detection self-efficacy emerged as a strong predictor of phishing susceptibility. For future research, it would be interesting to delve into how these individuals process/analyze phishing emails and to explore other characteristics that may influence their determination and certainty in not falling prey to phishing attempts. Using methodologies like Eye Tracker could provide valuable insights into the visual patterns used by individuals when examining phishing emails. For instance, are there different patterns in the exploration of emails, and if so, what visual patterns enable accurate detection of phishing attempts? Such investigations could further enhance our understanding of phishing susceptibility.

In the present study, we sought to overcome some limitations of previous research, such as email diversity. In contrast to former studies (e.g., Downs et al., 2007; Abroshan et al., 2020), we used four emails, while others used only two (one phishing and one legitimate email). Nevertheless, we believe that in future studies it is relevant to increase the number and diversity of phishing emails even further. We consider these improvements necessary, considering the studies by Ferreira and Lenzini (2015a; 2015b) which highlight the importance of different email characteristics based on principles such as distraction and authority in phishing susceptibility. Moreover, even though we were able to gather data from 449 individuals, the form of recruitment involved university staff, students, and other participants recruited through social networks. In future studies, it will be important to consider using other forms of recruitment to achieve a more diverse and representative sample. Additionally, it will be central to understand the percentage of participants recruited from different channels, which was not possible in the present work. In fact, we were unable to ascertain the precise percentage of participants obtained from each source. Consequently, we acknowledge that the potential variability in the proportion of participants from each source may introduce a menace to generalizability of the findings. Moreover, the device used to answer the survey (e.g., computer or smartphone) was not taken into consideration, which could be important to explore how different devices may influence the responses and, consequently, phishing susceptibility. It is also relevant to acknowledge the limitation regarding our sample size, especially the small portion of the subset of susceptible individuals. Even though the overall sample is considered adequate for the study, the limited sample size of the susceptible group may have implications for the generalizability of the findings and the statistical power to detect significant effects. Consequently, the results should be interpreted with caution, and the implications need to be considered in light of this constrain. To address this limitation, future research should aim to include larger and diverse sample in order to achieve a higher number of susceptible individuals to phishing.

Finally, our study contributes to an ongoing effort to develop a profile of individuals susceptible to phishing. Taking our findings into consideration, we emphasize the importance of training or educational programs focused on cognitive processing and self-efficacy phishing detection. Specifically, such training should provide individuals with knowledge about the cues they need to pay attention to when analyzing

emails to avoid failing for phishing attempts. Additionally, it should include tools to enhance individuals' self-efficacy in detecting phishing emails, protecting them for these attacks. Furthermore, targeted awareness campaigns could be developed, focusing on increasing individual's phishing detection self-efficacy and promoting safe online behaviors while conducting sensitive activities like home banking or financial transactions. Also important is tailoring these campaigns to various user groups based on their levels of self-efficacy. Lastly, the interplay between industries, institutions and cybersecurity experts is fundamental to share practices and combat phishing threats. These joint initiatives would be crucial to raise awareness among individuals and promote preventive measures. We believe that this approach can significantly improve the prevention of phishing susceptibility and, consequently, reduce the risk of falling victim to phishing attacks.

CRedit authorship contribution statement

Liliana Ribeiro: Writing – original draft, Conceptualization,

Appendix

A: Legitimate E-mail

↩ Responder ↩ Responder a todos → Reencaminhar ...

Ascendi - Pagamento nº d91b7 | Payment no d91b7

**Portal Ascendi** <noreply@ascendi.pt>
13/08/2020 08:36



Para: rcoutinho42i@hotmail.com

 ASCENDI2020081308351331730...
103,03 KB

Estimado(a) Cliente,

Enviamos-lhe a(s) sua(s) fatura(s)/recibo(s)
Na sequência do pagamento nº d91b7, no valor total de 8,52 € que efetuou no nosso Portal, enviamos em anexo a(s) fatura(s)/recibo(s) referente(s) ao mesmo.

Estamos ao seu dispor para qualquer esclarecimento.
Se necessitar de falar connosco, estamos disponíveis 24 horas por dia através do nosso [site](#). Em alternativa, pode usar a nossa linha de apoio a clientes (229 980 200).

Com os nossos cumprimentos,

Apoio a Clientes
Ascendi

Este é um e-mail gerado automaticamente. Por favor não responda para este endereço.

B: Phishing E-mail

Methodology, Investigation, Resources, Formal analysis, Writing – review & editing. **Inês Sousa Guedes:** Formal analysis, Data curation, Writing – review & editing, Supervision. **Carla Sofia Cardoso:** Formal analysis, Data curation, Visualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

↩ Responder ↩ Responder a todos → Reencaminhar 📁 Arquivar

A sua fatura EDP (contrato 298489271582)



Para:josef14r@hotmail.com

Edp serviço universal

Estimado (a) Cliente,

após as últimas cálculos das suas contas,

determinamos que você é elegível para receber um reembolso de 52,56 €

Por favor, envie sua solicitação de reembolso para que possamos tratá-lo o mais rápido possível

É rápido e fácil, por favor

Consulte os procedimentos a seguir

Por favor note que o seu pedido será aprovado dentro de 15 dias úteis a partir da data do pedido.
graças à sua conta, você pode sempre com segurança:

- ☐ acessar suas contas
- ☐ manter um olho sobre o seu consumo

C: Legitimate E-mail

↩ Responder ↩ Responder a todos → Reencaminhar ...

Encomenda DA533269200PT



Para: filpintorq@hotmail.com

A sua encomenda [DA533269200PT](#) de BERTRAND.PT será entregue no dia 23/03, até às 19h Para alterar aceda a www.ctt.pt/alterar_encomenda

Detalhe da Encomenda

Nome destinatário: JOANA VIEIRA

Morada: RUA DA ESTAÇÃO, 947

Código Postal: 4430-975

Obrigado pela sua preferência.

CTT

www.ctt.pt

Esta mensagem é enviada automaticamente, por favor não responda.

Em caso de dúvidas ou informações adicionais, aceda a www.ctt.pt/ajuda

D: Phishing E-mail

↶ Responder ↶ Responder a todos → Reencaminhar ...

VOCÊ TEM UMA PACOTE

CTT Correios de Portugal <info@ctt.pw>

24/03/2014 13:45

Para: "(Nome)"

A empresa de courier nao foi capaz de entergar o seu EA42554294PT pacote para o seu endereco.

Causa: Erro no endereco de entrega.

Faca o download das informacoes sobre pacote no site da CTT Expresso, imprimi-lo e ir para correios para receber sua pacote.

DOWNLOAD DE INFORMAÇÕES

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Liliana Ribeiro: Degree in Criminology from the Faculty of Law of the University of Porto (FDUP). Currently, Master's student in the same Faculty. Develops research in the areas of security and cybercrime.

Inês Sousa Guedes: Master and Ph.D. in Criminology from the Faculty of Law of the University of Porto (FDUP). Assistant Professor at the same University. She is a founding and integrated member of the Center for Interdisciplinary Research-Crime, Justice, and Security (CJS) of the FDUP and a collaborating member of the Center for Legal, Economic, and Environmental Studies (CEJEA).

Carla Sofia Cardoso: Ph.D. in Biomedical Sciences from the Institute of Biomedical Sciences Abel Salazar of the University of Porto. Assistant Professor at the Faculty of Law of the University of Porto (FDUP), teaches in all three cycles of studies in Criminology. She is a founding and integrated member of the Center for Interdisciplinary Research-Crime, Justice, and Security (CJS) of the FDUP.